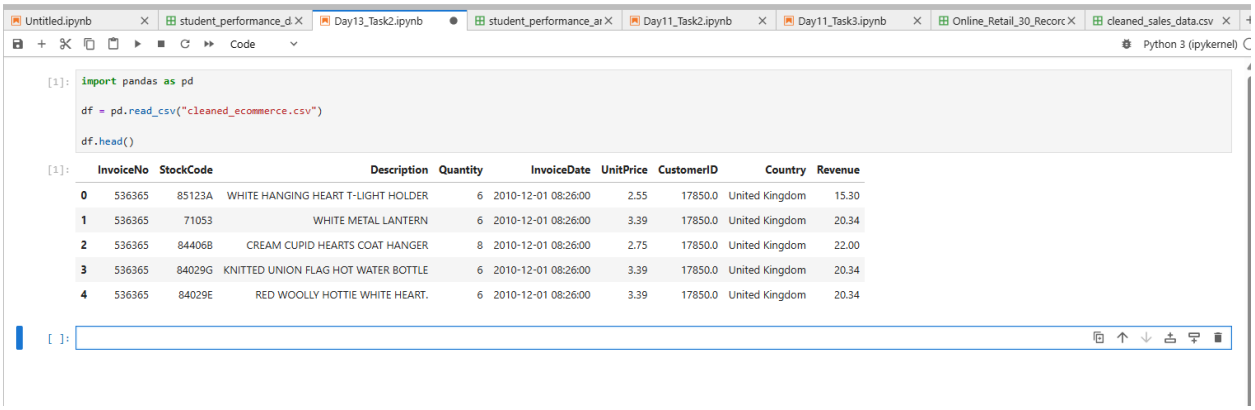
Day 13 Task 2

Create the DataFrame



Untitled.ipynb | student_performance_d | Day13_Task2.ipynb | student_performance_ar | Day11_Task2.ipynb | Day11_Task3.ipynb | Online_Retail_30_Recorc | cleaned_sales_data.csv

Code | Python 3 (ipykernel)

```
[1]: import pandas as pd

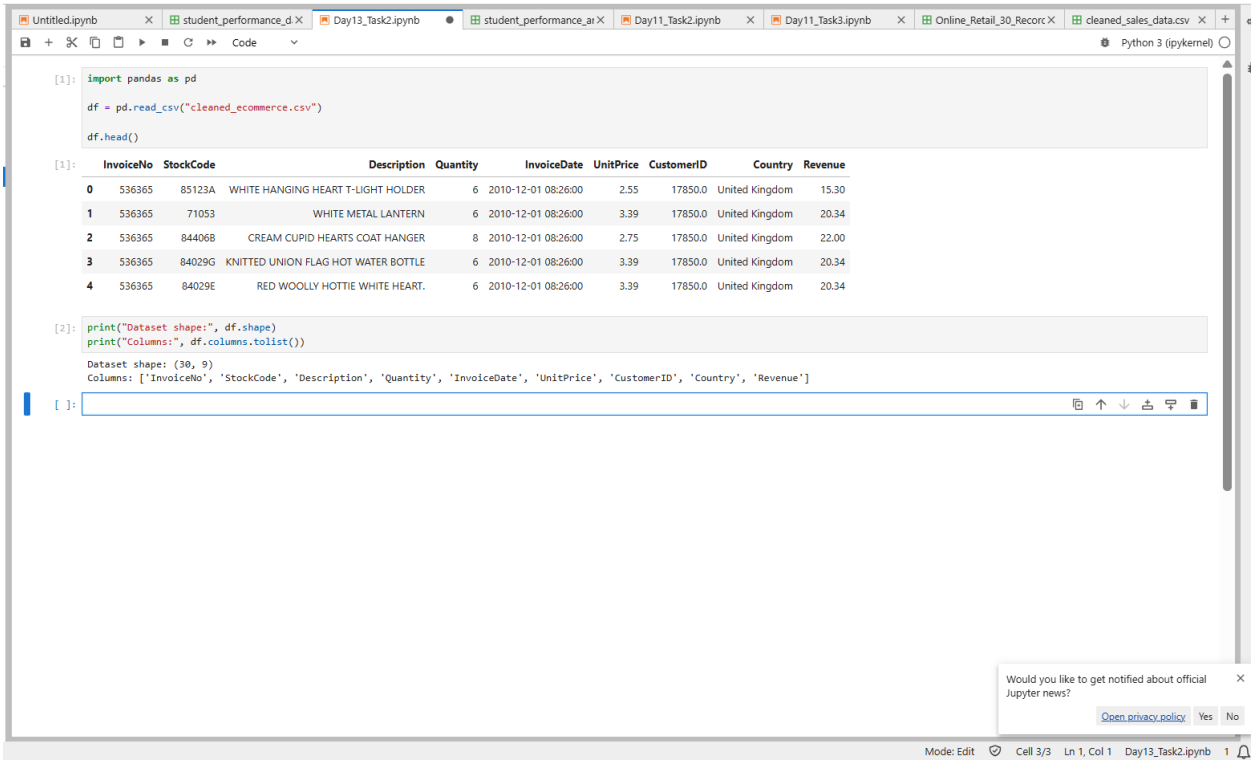
     df = pd.read_csv("cleaned_ecommerce.csv")

     df.head()
```

[1]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

```
[ ]:
```



Untitled.ipynb | student_performance_d | Day13_Task2.ipynb | student_performance_ar | Day11_Task2.ipynb | Day11_Task3.ipynb | Online_Retail_30_Recorc | cleaned_sales_data.csv

Code | Python 3 (ipykernel)

```
[1]: import pandas as pd

     df = pd.read_csv("cleaned_ecommerce.csv")

     df.head()
```

[1]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

```
[2]: print("Dataset shape:", df.shape)
     print("Columns:", df.columns.tolist())
```

```
Dataset shape: (30, 9)
Columns: ['InvoiceNo', 'StockCode', 'Description', 'Quantity', 'InvoiceDate', 'UnitPrice', 'CustomerID', 'Country', 'Revenue']
```

```
[ ]:
```

Would you like to get notified about official Jupyter news?

Open privacy policy | Yes | No

Mode: Edit | Cell 3/3 | Ln 1, Col 1 | Day13_Task2.ipynb | 1

Check Unique Orders and Row Count

```
[3]: print("Unique Orders:", df["InvoiceNo"].nunique())
     print("Total Rows:", len(df))
```

```
Unique Orders: 6
Total Rows: 30
```

```
[ ]:
```

Import Libraries and Convert Dates

```
[5]: import numpy as np
     import matplotlib.pyplot as plt

     df["InvoiceDate"] = pd.to_datetime(
         df["InvoiceDate"],
         errors="coerce"
     )

     print(df["InvoiceDate"].isnull().sum())
     print(df.dtypes)
```

```
0
InvoiceNo               int64
StockCode                 str
Description               str
Quantity                int64
InvoiceDate    datetime64[us]
UnitPrice             float64
CustomerID            float64
Country                   str
Revenue               float64
dtype: object
```

```
[ ]:
```

Create the Revenue Column

```
[6]: df["Quantity"] = pd.to_numeric(
         df["Quantity"], errors="coerce"
     )

     df["UnitPrice"] = pd.to_numeric(
         df["UnitPrice"], errors="coerce"
     )

     df["Revenue"] = (
         df["Quantity"] * df["UnitPrice"]
     )

     df.head()
```

[6]:

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country	Revenue
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom	15.30
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom	22.00
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom	20.34

```
[ ]:
```

Find the five highest-revenue records

[7]:
```
highest_revenue_orders = (
    df.sort_values(
        "Revenue",
        ascending=False
    )[
        [
            "InvoiceNo",
            "InvoiceDate",
            "Country",
            "Description",
            "Quantity",
            "UnitPrice",
            "Revenue"
        ]
    ]
    .head(5)
)

highest_revenue_orders
```

[7]:

	InvoiceNo	InvoiceDate	Country	Description	Quantity	UnitPrice	Revenue
27	536370	2010-12-01 08:45:00	France	ALARM CLOCK BAKELIKE RED	24	3.75	90.00
26	536370	2010-12-01 08:45:00	France	ALARM CLOCK BAKELIKE PINK	24	3.75	90.00
9	536367	2010-12-01 08:34:00	United Kingdom	ASSORTED COLOUR BIRD ORNAMENT	32	1.69	54.08
28	536370	2010-12-01 08:45:00	France	ALARM CLOCK BAKELIKE GREEN	12	3.75	45.00
20	536367	2010-12-01 08:34:00	United Kingdom	DOORMAT NEW ENGLAND	4	7.95	31.80

[ ]:

Create a country summary

[8]:
```
country_summary = (
    df.groupby("Country")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        Orders=("InvoiceNo", "nunique"),
        AverageRevenue=("Revenue", "mean")
    )
    .sort_values(
        "TotalRevenue",
        ascending=False
    )
)

country_summary
```

[8]:

Country	TotalRevenue	Orders	AverageRevenue
United Kingdom	527.95	5	20.305769
France	235.20	1	58.800000

[ ]:

Create a product summary

[9]:
```
product_summary = (
    df.groupby("Description")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        UnitsSold=("Quantity", "sum"),
        Orders=("InvoiceNo", "nunique")
    )
    .sort_values(
        "TotalRevenue",
        ascending=False
    )
)

product_summary
```

[9]:

Description	TotalRevenue	UnitsSold	Orders
ALARM CLOCK BAKELIKE PINK	90.00	24	1
ALARM CLOCK BAKELIKE RED	90.00	24	1
ASSORTED COLOUR BIRD ORNAMENT	54.08	32	1
ALARM CLOCK BAKELIKE GREEN	45.00	12	1
DOORMAT NEW ENGLAND	31.80	4	1
RECIPE BOX WITH METAL HEART	31.80	4	1
FELTCRAFT PRINCESS CHARLOTTE DOLL	30.00	8	1
BOX OF 6 ASSORTED COLOUR TEASPOONS	25.50	6	1
JAM MAKING SET WITH JARS	25.50	6	1
GLASS STAR FROSTED T-LIGHT HOLDER	25.50	6	1
CREAM CUPID HEARTS COAT HANGER	22.00	8	1
RED WOOLLY HOTTIE WHITE HEART.	20.34	6	1
KNITTED UNION FLAG HOT WATER BOTTLE	20.34	6	1
WHITE METAL LANTERN	20.34	6	1
BOX OF VINTAGE ALPHABET BLOCKS	19.90	2	1
LOVE BUILDING BLOCK WORD	17.85	3	1
HOME BUILDING BLOCK WORD	17.85	3	1
BATH BUILDING BLOCK WORD	17.85	3	1
SET 7 BABUSHKA NESTING BOXES	15.30	2	1
WHITE HANGING HEART T-LIGHT HOLDER	15.30	6	1
BLUE COAT RACK PARIS FASHION	14.85	3	1
BOX OF VINTAGE JIGSAW BLOCKS	14.85	3	1

Would you like to get notified about official Jupyter news?
Open privacy policy Yes

Create a monthly summary

[10]:
```
df["Month"] = df["InvoiceDate"].dt.to_period("M")

monthly_summary = (
    df.groupby("Month")
    .agg(
        TotalRevenue=("Revenue", "sum"),
        Orders=("InvoiceNo", "nunique")
    )
    .sort_index()
)

monthly_summary
```

[10]:

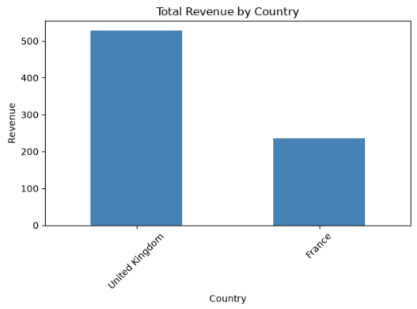
Month	TotalRevenue	Orders
2010-12	763.15	6

[ ]:

Plot total revenue by country

[11]:
```
country_summary["TotalRevenue"].head(10).plot(
    kind="bar",
    color="steelblue",
    title="Total Revenue by Country"
)

plt.xlabel("Country")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Would you like to get notified about official

Plot total revenue by product

[12]:
```
product_summary["TotalRevenue"].head(10).plot(
    kind="bar",
    color="darkorange",
    title="Total Revenue by Product"
)

plt.xlabel("Product")
plt.ylabel("Revenue")
plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```

